

Experiment 4

Electronic Codebook (ECB) Mode

Aim

To implement the Electronic Codebook (ECB) mode of DES for encrypting and decrypting plaintext.

Description

Electronic Codebook (ECB) is the simplest mode of operation in DES. In this mode, the plaintext is divided into fixed-size 64-bit blocks, and each block is encrypted independently using the same secret key. Since identical plaintext blocks produce identical ciphertext blocks, ECB is easy to implement but is considered less secure for encrypting large amounts of data.

Algorithm

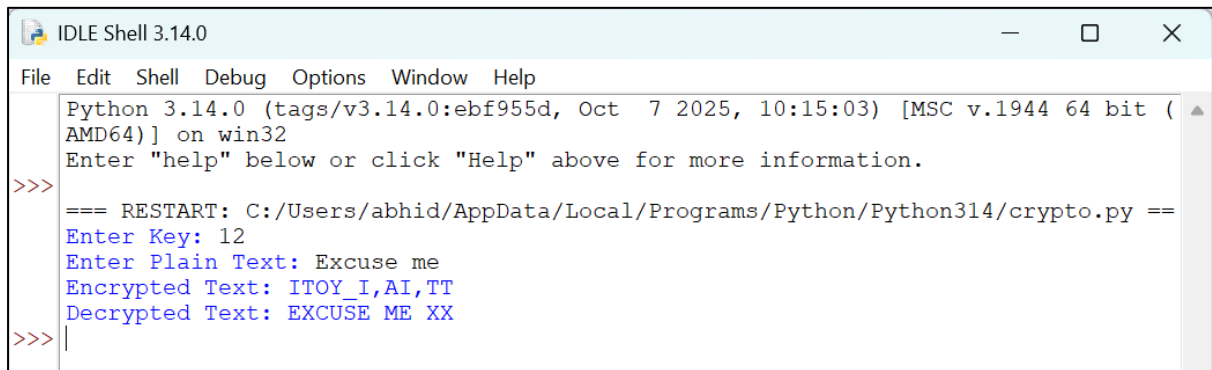
1. Start the program.
2. Read the plaintext from the user.
3. Divide the plaintext into fixed-size blocks.
4. Encrypt each block independently using the same key.
5. Combine the encrypted blocks to form the ciphertext.
6. Display the encrypted text.
7. Decrypt each ciphertext block independently using the same key.
8. Combine the decrypted blocks to obtain the original plaintext.
9. Display the decrypted text.
10. Stop the program.

Program:

```
key = int(input("Enter Key: "))
plaintext = input("Enter Plain Text: ").upper()
while len(plaintext) % 4 != 0:
    plaintext += "X"
encrypted = ""
```

```
for i in range(0, len(plaintext), 4):  
    block = plaintext[i:i+4]  
    for ch in block:  
        encrypted += chr((ord(ch) ^ key) % 128)  
print("Encrypted Text:", encrypted)  
decrypted = ""  
for ch in encrypted:  
    decrypted += chr((ord(ch) ^ key) % 128)  
print("Decrypted Text:", decrypted)
```

Output:



```
IDLE Shell 3.14.0  
File Edit Shell Debug Options Window Help  
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>> === RESTART: C:/Users/abhid/AppData/Local/Programs/Python/Python314/crypto.py ==  
Enter Key: 12  
Enter Plain Text: Excuse me  
Encrypted Text: ITOY_I,AI,TT  
Decrypted Text: EXCUSE ME XX  
>>> |
```